

Tuning ArduPilot Speed and Heading Control for Small Unmanned Surface Vehicles: A Systematic Approach Grounded in Classical Control Theory

1. Problem

The Navy increasingly relies on small unmanned surface vehicles (USVs), many of which run ArduPilot Rover configured through Mission Planner. Rover's tuning guidance is shared with wheeled ground vehicles and does not state which of its several hundred parameters matter most on water, which ones overlap, or how each one relates to classical control. In practice, parameters are set by trial and error, which costs water time and can produce unsafe or poorly performing vehicles.

2. Research questions

1. Which ArduPilot Rover parameters most affect the speed and heading response of a small USV?
2. How can those parameters be set to achieve the desired response with minimal control effort, reducing actuator wear and extending endurance?
3. Which parameters should a user prioritize when tuning a small USV, and in what order?
4. Which parameters are redundant, either duplicating another parameter or always overridden by a tighter limit, and which have no measurable effect on a boat?
5. Which classical control element does each important parameter implement, and does it behave as control theory predicts?
6. Do the parameter rankings found in simulation hold on the real autopilot hardware and on the water?
7. Is a simple linear boat model good enough for tuning ArduPilot, or does a nonlinear model give measurably better performance on the water?

3. Expected contributions

- A prioritized set of ArduPilot Rover parameters for small USVs, with a recommended tuning order and an identification of redundant and ineffective parameters.
- A correspondence between ArduPilot parameters and the classical control elements they implement, verified against the ArduPilot source code and tested in simulation and on the water.
- Guidance on setting the turning limits of a small USV to carry speed through turns without capsizing.
- Open-source tools for model identification, simulation, and automated testing, including a working hardware-in-the-loop setup for ArduPilot boats, that allow other users to repeat this process on their own vessels.
- A quantitative comparison of SITL, HIL, and on-water results, showing how well simulation predicts both the parameter rankings and the speed and heading response of the real boat, and how much tuning with a nonlinear rather than a linear boat model improves waypoint performance.

5. Research plan

